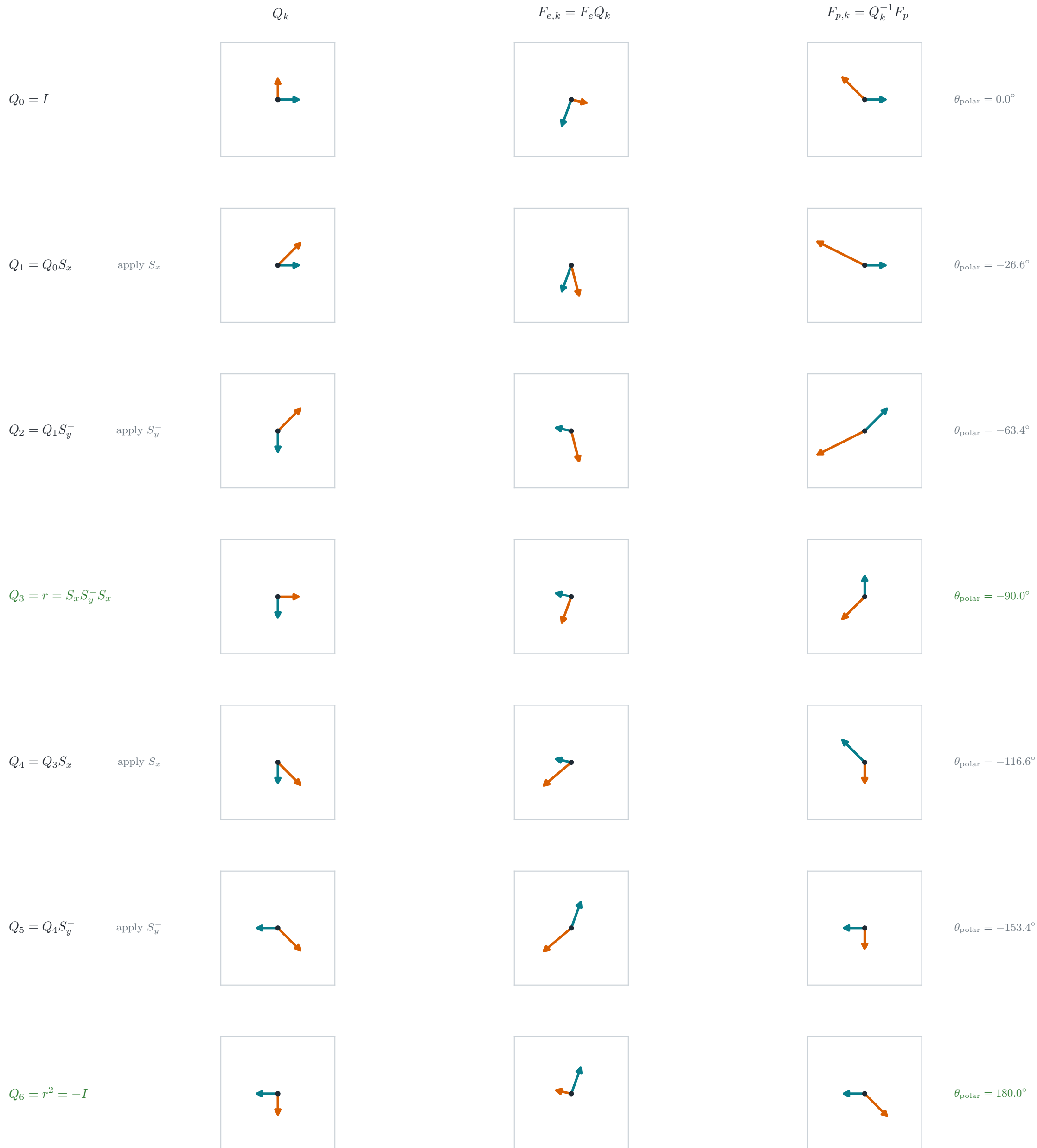


Six unit shears generate $r^2 = -I$

$$F_{e,k} = F_e Q_k, F_{p,k} = Q_k^{-1} F_p, F = F_{e,k} F_{p,k}, \det Q_k = +1$$

Only $Q_3 = r$ and $Q_6 = r^2$ are pure orthogonal rotations; the other angles are polar-shear rotations.



At $k = 6$: $(F_e Q_6, Q_6^{-1} F_p) = (-F_e, -F_p)$, so F and both metric tensors are unchanged.